FPGA Synthesis and Implementation Report: ECG Arrhythmia Classifier

Oct 9, 2026 · S Harish Adithya, Rajalakshmi Engineering College

The ECG classifier passes Vivado synthesis and place-and-route for the PYNQ's Zynq-7020 chip at 50 MHz with timing met (setup slack +5.42 ns, hold slack +0.11 ns). It uses 394 LUTs, 213 flip-flops, 17.5 block-RAM tiles and 2 DSP slices, and classifies one heartbeat in 2.59 ms.

Results at a glance

Place-and-route confirmed the synthesis estimates: the real wire delays cost about 1.2 ns of slack, and resource use did not change.

Metric	After synthesis (estimate)	After place and route (real)
Setup slack at 20 ns clock	+6.622 ns	**+5.424 ns**
Hold slack	not checked at this stage	**+0.108 ns**
Slowest path	13.05 ns	**14.05 ns**
Rough maximum clock	about 75 MHz	**about 69 MHz**
LUTs	398 (0.75%)	**394 (0.74%)**
Flip-flops	213 (0.20%)	**213 (0.20%)**
Block RAM tiles	17.5 (12.5%)	**17.5 (12.5%)**
DSP slices	2 (0.91%)	**2 (0.91%)**
Slices	not counted	**128 (0.96%)**
Estimated on-chip power	not run	**0.126 W** (0.022 W dynamic)
Errors / critical warnings	0 / 0	**0 / 0**

Rough maximum clock is 1000 / (20 ns minus the slack), an estimate and not a tested limit. Percentages are of the xc7z020 chip.

Design and flow

The whole classifier is one Verilog module, `ecg_top`. One shared multiply-accumulate unit runs conv1, conv2, fc1 and fc2 one after another, which keeps the logic small.

Item	Value
Tool	Vivado 2026.1, Tcl scripts `run_synth.tcl` and `run_impl.tcl`
Chip	xc7z020clg400-1, the Zynq-7020 on the PYNQ-Z1 and PYNQ-Z2 boards
Clock	50 MHz (20 ns) constraint on `clk`
Mode	Out of context: the classifier alone, with no pins, board or processor around it
Steps	`synth_design`, then `opt_design`, `place_design`, `phys_opt_design`, `route_design` (default settings)
Arithmetic	16-bit fixed point; 16 × 16 multiply into a 48-bit accumulator
Memory	32,488 weights × 16 bit (about 63 KB) in block RAM, 58 biases, four activation buffers
Accuracy	98.51% on 24,526 beats from 10 held-out patients (Python fixed-point model); the hardware matched that model exactly, value by value, on 64 test beats

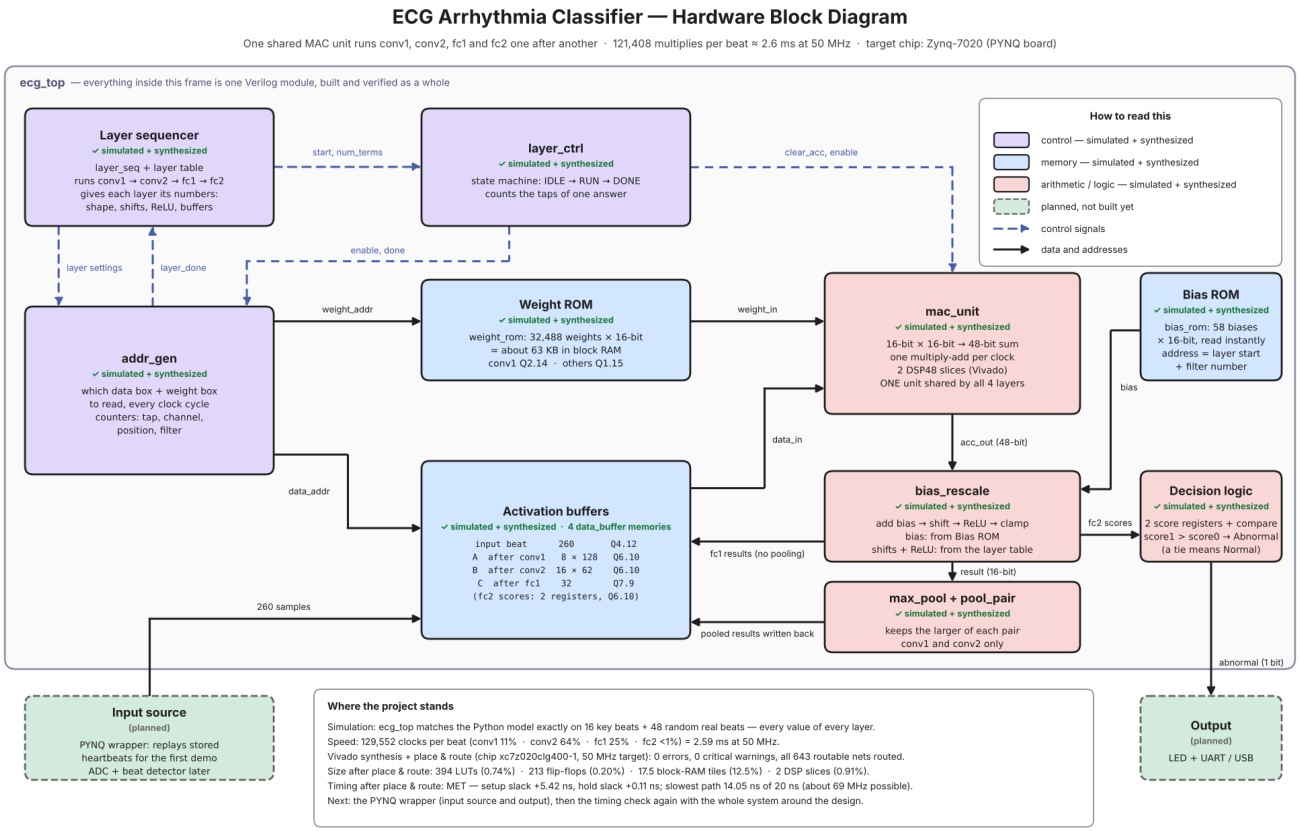


Figure 1. Block diagram of the classifier, with the post-route results.

Timing after place and route

All timing checks pass at the 50 MHz target: worst setup slack is +5.424 ns, worst hold slack is +0.108 ns, and none of the 877 checked endpoints fails.

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS)	5.424 ns	Worst Hold Slack (WHS)	0.108 ns	Worst Pulse Width Slack (WPWS)	8.750 ns
Total Negative Slack (TNS)	0.000 ns	Total Hold Slack (THS)	0.000 ns	Total Pulse Width Negative Slack (TPWS)	0.000 ns
Number of Failing Endpoints	0	Number of Failing Endpoints	0	Number of Failing Endpoints	0
Total Number of Endpoints	877	Total Number of Endpoints	877	Total Number of Endpoints	259

All user specified timing constraints are met.

Figure 2. Design Timing Summary after place and route.

- **Setup** slack is how much earlier than the 20 ns deadline the slowest signal arrives. The synthesis estimate was +6.622 ns, so the real wiring cost about 1.2 ns.

- **Hold** slack (+0.108 ns) checks that signals do not change too soon after the clock edge. It is positive on all 877 endpoints.
- **Pulse width** slack (+8.750 ns, 259 endpoints) checks the clock high and low times at the block RAMs and DSPs. None fails.

Slowest path

The slowest path takes 14.049 ns of the 20 ns clock. It runs from the layer-number register through the bias-and-rescale step and the max-pool compare into the `buf_b` block RAM.

Name	Slack ¹	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Logic %	Net %	Requirement	Source Clock
Constrained Paths (1)												
Path 1	5.424	20	143	seq/layer_r_reg[0]/C	buf_b/mem_reg/DIADI[0]	14.049	5.319	8.730	37.9	62.1	20.000	clk

Figure 3. The slowest setup path after routing (Vivado timing report).

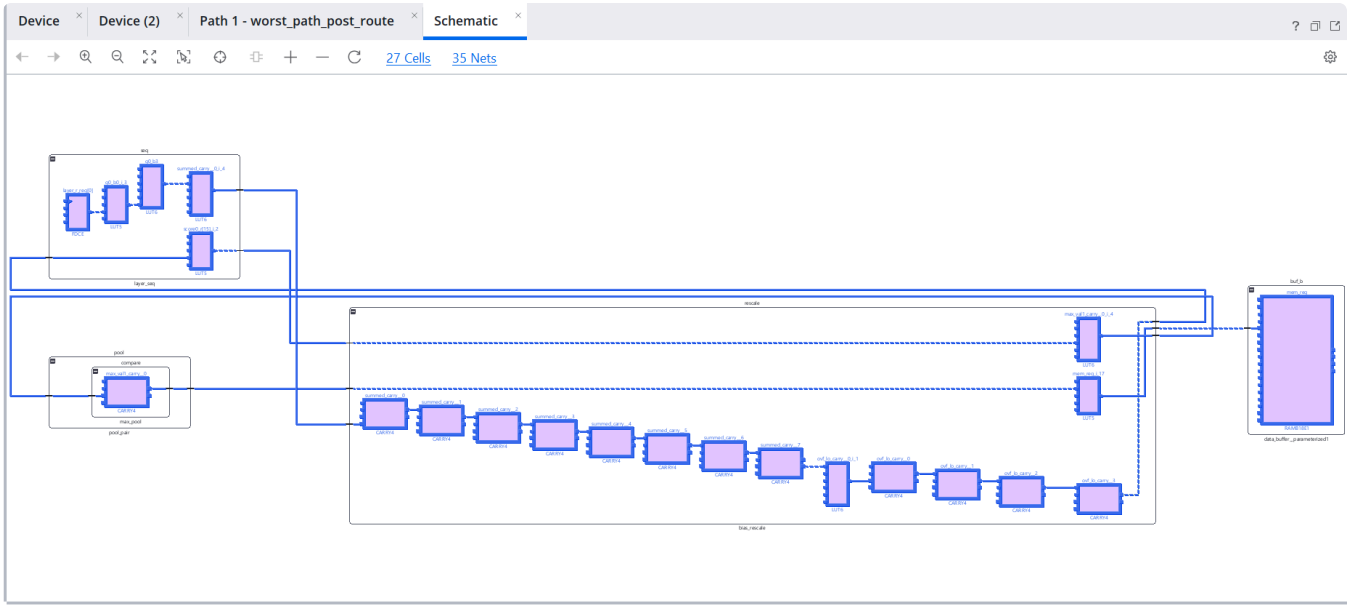


Figure 4. Schematic of the same path: layer-settings logic, bias-add and clamp carry chains, max-pool compare, block RAM write.

- 20 logic levels: logic 5.319 ns (37.9%) and wires 8.730 ns (62.1%).
- The highest fanout on the path is 143 loads, most likely the layer-number signal that selects each layer's settings.
- Clock skew is −0.049 ns and clock uncertainty 0.035 ns.

- No change is needed at 50 MHz. For a faster clock, the first options are duplicating the layer register or adding a register stage after the rescale step; the second would need re-verification.

Resource use

The classifier uses under 1% of the chip's logic and 12.5% of its block RAM: 394 LUTs, 213 flip-flops, 128 slices, 17.5 block-RAM tiles and 2 DSP slices.

Summary

Resource	Utilization	Available	Utilization %
LUT	394	53200	0.74
LUTRAM	12	17400	0.07
FF	213	106400	0.20
BRAM	17.50	140	12.50
DSP	2	220	0.91

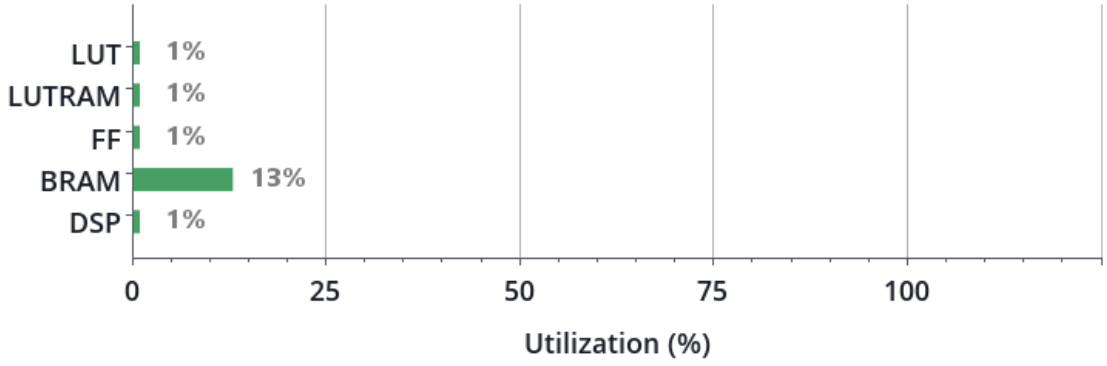


Figure 5. Whole-design utilization after place and route (xc7z020).

Hierarchy

Name	Slice LUTs (53200)	Slice Registers (106400)	Slice (13300)	LUT as Logic (53200)	LUT as Memory (17400)	Block RAM Tile (140)	DSPs (220)
ecg_top	394	213	128	382	12	17.5	2
agen (addr_gen)	81	62	35	81	0	0	0
mac (mac_unit)	80	48	24	80	0	0	2
rescale (bias_rescale)	106	0	49	106	0	0	0
seq (layer_seq)	78	5	32	78	0	0	0

Figure 6. Utilization by module after place and route.

- The 12 LUTs used as memory are part of the 394 LUTs, not extra.
- `mac_unit` holds both DSP slices. `bias_rescale` is the largest logic block at 106 LUTs.
- Block RAM shows on the top row because the memories are merged into `ecg_top`. It is mostly the 32,488 16-bit weights (about 63 KB).

Placement and routing

The design fits in the bottom two clock regions (X0Y0 and X1Y0) of the programmable logic, and all 643 routable nets are fully routed with 0 routing errors.

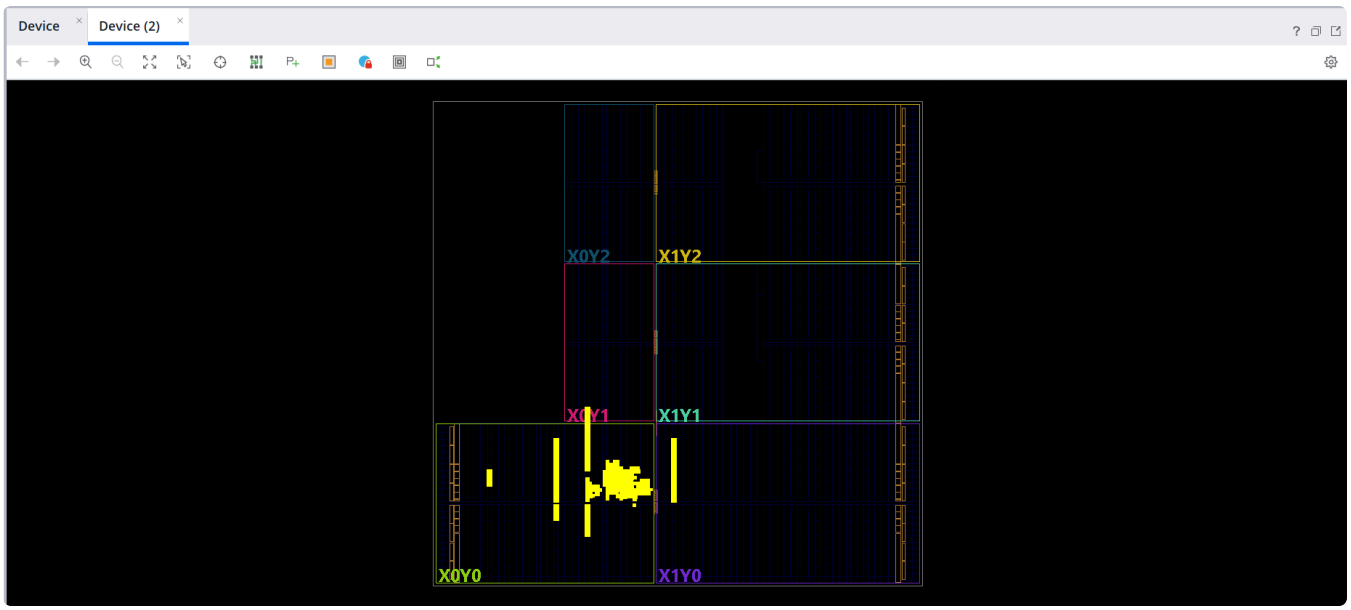


Figure 7. Placement of the classifier (yellow) on the xc7z020. The thin vertical bars are block RAM columns; the black area at the top left is the processor region.

- Of 980 logical nets, 337 need no routing: 297 are inside blocks, 8 have no loads and 32 are ports.
- The other 643 nets are routed, with 0 routing errors.

Power estimate

Vivado estimates 0.126 W total on-chip power at 50 MHz: 0.104 W is the chip's own static power and 0.022 W is the design's dynamic power.

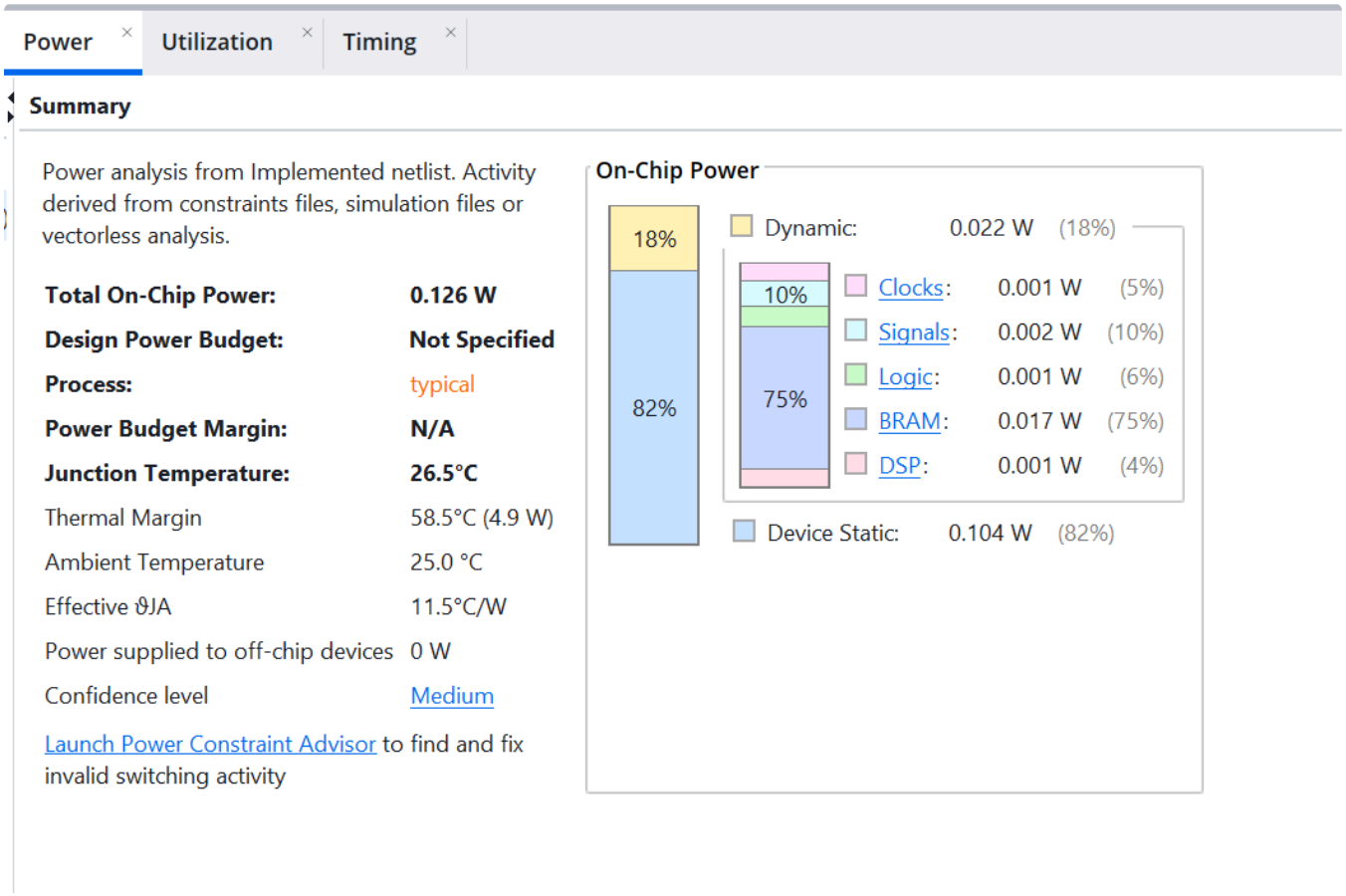


Figure 8. Vivado power estimate after place and route (confidence: medium).

- Block RAM accounts for 75% of the dynamic power (0.017 W). Signals take 0.002 W; clocks, logic and DSP take 0.001 W each.
- Rough energy per beat is 22 mW × 2.59 ms ≈ 57 µJ, dynamic part only.
- Junction temperature is 26.5 °C at 25 °C ambient.
- This is a vectorless estimate: Vivado assumes default switching activity, not real heartbeat data. It covers the classifier only and leaves out the Zynq's ARM processor, the board and the pins. Power measured on the board is still to do.

Speed per beat

One beat takes exactly 129,552 clocks, which is 2.59 ms at 50 MHz and about 116 times faster than the 300 ms real-time budget (one beat per 300 ms, or 200 beats per minute).

Layer	Multiplications	Clocks	Share of beat
conv1	10,240	14,338	11%
conv2	79,360	83,330	64%
fc1	31,744	31,810	25%
fc2	64	70	under 1%
Start and layer changes	0	4	under 1%
Total	**121,408**	**129,552**	**100%**

The single multiplier does useful work on 94% of the clocks. The rest is 2 clocks of pipeline overhead per output, plus the 4 clocks between layers.

Throughput: run back to back, the design handles about 385 beats per second (50 MHz divided by 129,812 clocks: 129,552 of work plus 260 to load each beat), or about 47 million multiply-accumulates per second. A heart at 200 beats per minute produces 3.3 beats per second, so the classifier has about 116 times more capacity than one patient needs. This is the design alone; the PYNQ wrapper's loading speed may lower it.

Comparison with published work

Among the published FPGA ECG classifiers reviewed here that report resources, this design has the smallest logic footprint. Accuracy cannot be ranked directly, because the task and the test split differ.

Work	Platform	Task and test split	Accuracy	LUT / FF / BRAM / DSP	Latency per beat	Power
This work	Zynq-7020, 50 MHz (classifier only)	2 classes, inter-patient (10 held-out records)	98.51%	**394 / 213 / 17.5 / 2**	2.59 ms	0.126 W (estimate)

Work	Platform	Task and test split	Accuracy	LUT / FF / BRAM / DSP	Latency per beat	Power
ECG-LDC 2026	PYNQ-Z2, 50 MHz	5 AAMI classes, random 70/30 split	97.18%	core 5,652 / 4,790 / 4 / 0; full system 8,255 / 8,296 / 6 / 0	23.9 µs core; 4.2 ms end to end	1.71 W (full system)
Fradi 2024	PYNQ-Z2 (HLS)	MIT-BIH, split not stated	98.76–99.03%	1,477 / 2,191 / not stated / 3	about 2 ms (the paper's figures disagree)	not stated
Liu 2025 †	Zynq-7Z020, 50 MHz	5 classes	96.55%	13,726 / 18,379 / 5.5 / 9	63 ms	1.78 W
Greco 2025 †	Zybo Z7-20, 50 MHz	5 classes	91.90%	20,198 / 12,933 / 50.5 / 98	not stated	1.80 W
Chen 2021 †	Artix-7, 100 MHz	2 classes	93.60%	5,981 / 3,128 / not stated / 53	0.257 ms	not stated
Quantized NN accelerator (Semnan journal)	Artix-7, 100 MHz	3 classes (N, V, S), inter-patient	98.4%	not stated	8.5 µs	0.92 W

† Values as tabulated in the ECG-LDC paper, not checked against the original papers.

What is and is not comparable

- **Task.** Ours is Normal vs Abnormal, with the A, V, L and R beat types merged. The others use 2, 3 or 5 classes, and a 2-class accuracy is not comparable with a 5-class one.
- **Test split.** A random split lets the same patient appear in training and test, which inflates accuracy. A 2025 review reports drops of about 11 and 6 points when two studies moved to inter-patient testing. Ours is inter-patient, but on our own 10 test records, not the standard published split.
- **Resources.** Ours counts the classifier alone. Others count a full system with processor interfaces: ECG-LDC uses 5,652 LUTs for its core but 8,255 for the full system.
- **Power.** Ours is an estimate for the classifier alone. The others include the processor system.

Where this design stands

- **Strongest:** 3.7 to 51 times fewer LUTs than the FPGA designs above that report them, and 2 DSP slices where the others use 0 to 98.
- **Weaker:** speed, at 2.59 ms and about 385 beats per second, against about 24,400 beats per second for ECG-LDC's core and about 118,000 for the Semnan design, though still about 116 times inside the real-time budget. Block RAM is 17.5 tiles against 4 to 6 for ECG-LDC, because the weights are stored uncompressed at 16 bits (ECG-LDC's model is 3.4 kB). The task is binary only.
- **Energy per beat:** the review quotes 2.05 µJ (Zhang, 1-D CNN on a ZC706) and 346 µJ (Xing, spiking network). Ours is about 57 µJ, dynamic part only and estimated.
- **For a fair comparison:** test on the standard split with per-class sensitivity, and measure power on the board.

Limitations and next steps

The results are for the classifier on its own. These points limit what they prove:

- **Timed alone.** The design was implemented with an ideal clock and no pins, so paths from the input pins and to the outputs were not timed, and Vivado ran only part of its design-rule checks.
- **No bitstream yet.** Making one needs the PYNQ wrapper: clock and reset from the processor, and a way to load the beat.
- **Verification.** Simulation used Icarus Verilog (exact match with the Python model on 64 beats) plus a gate-level check in Yosys. Vivado's own simulator could not run the testbench on the development PC, where it hung at start-up, and the routed netlist has not been simulated.
- **Power** is a Vivado estimate, not a measurement.
- **Task scope.** Normal vs Abnormal on five beat types, not the full five-class AAMI task.
- **Reset** is asynchronous, so the wrapper should supply a synchronized reset.

Next steps, in order:

1. Build the PYNQ wrapper that replays stored beats through `ecg_top`.
2. Re-run implementation with the wrapper for full-system timing and a bitstream.
3. Run on the board and measure power and time per beat.
4. Test the model on the standard split with per-class results.
5. Later: live ECG input.

Sources

Papers opened on 9 October 2026. All figures for other work come from these pages.

- ECG-LDC: A Hardware-Efficient Low-Dimensional Computing Framework for ECG Arrhythmia Classification (Tran, Tran, Do; 2026 preprint). Source of the ECG-LDC figures and of the Liu, Greco and Chen rows.
- A Systematic Review of ECG Arrhythmia Classification: Adherence to Standards, Fair Evaluation, and Embedded Feasibility (Silva et al.; 2025). Source for random vs inter-patient splits and the Zhang and Xing energy figures.
- FPGA Implementation of a CNN Application for ECG Class Detection (Fradi et al.; Traitement du Signal, 2024).
- Low-Power FPGA-Based Quantized Neural Network Accelerator for Real-Time Arrhythmia Detection from Single-Lead ECG (Sharif University of Technology; J. Modeling and Simulation in Electrical and Electronics Engineering, 2026).
- Not used: Nandini and Seshikala 2025 reports 99.7% accuracy and 0.42 mW but states no dataset, classes, split or resources.